

CS2601 Linear and Convex Optimization

Homework 3

Due: 2023.10.31

1. Suppose $C \subset \mathbb{R}^n$ is a nonempty closed convex set. Let $\mathcal{P}_C$ the projection onto C . Show that for any $\mathbf{x} \notin C$ and $\mathbf{y} \in C$,

$$\|\mathcal{P}_C(\mathbf{x}) - \mathbf{y}\|_2 < \|\mathbf{x} - \mathbf{y}\|_2$$

Hint: It is useful to review the proof for the nonexpansiveness of the projection operator. To get some intuition, try to visualize it for the case where $n = 2$ and C is a halfspace.

2. Suppose f is a convex function and $S \subset \text{dom } f$ is a convex set. Let M be the set of global minima of f over S ,

$$M = \{\mathbf{x}^* \in S : f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in S\}.$$

Show that M is a convex set.

3. Determine if the following functions are convex, concave, or neither.

- (a). $f(\mathbf{x}) = f(x_1, x_2, x_3) = x_1^2 + x_1x_3 + x_2^2 + x_2x_3 + \frac{1}{2}x_3^2$ on $\mathbb{R}^3$
- (b). $f(\mathbf{x}) = f(x_1, x_2) = (x_1x_2)^{-1}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$
- (c). $f(x_1, x_2) = x_1x_2^2$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$
- (d). $f(x_1, x_2) = x_1x_2^{-1/2}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$
- (e). $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

4. Is the following set convex? Show your argument.

$$S = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : \max\{\|\mathbf{A}\mathbf{x} + \mathbf{b}\|^3, \log(1 + e^{3x_1+2x_2})\} \leq 2\}.$$

You can use any results we have proved in class.

5. Let $f : C \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function defined on a nonempty open convex set C . Show that f is convex if and only if

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0, \quad \forall \mathbf{x}, \mathbf{y} \in C. \quad (1)$$

Hint: For the sufficiency, consider the restrictions of f to straight lines, and note that a univariate function h is increasing iff $[h(t) - h(s)](t - s) \geq 0$. You can assume the fact that the intersection of C with a straight line is an open interval when it is not empty.